

Donation to Food Banks during Food Security Crises: Experiment on Information Nudges with Intertemporal Giving Choices

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ABSTRACT

This study examined the effects of nudging type (deliberate vs. impulsive) and donation timing (today vs. in a month) using an incentive-compatible food bank donation experiment during a food security crisis. Donation rates were higher under deliberate nudging than under impulsive nudging and higher when donations were scheduled for a month later rather than made immediately. Deliberate nudging was particularly effective among individuals with lower empathy or higher income, while impulsive nudging had a stronger impact on those with higher empathy. Individuals with lower time-discount rates were less likely to donate when the donation was immediate but were equally likely to donate when it was delayed. Furthermore, beneficiaries of food assistance programs showed higher donation rates than non-beneficiaries. These results suggest that deliberate nudging and delayed donation options can substantially enhance charitable giving, especially among less empathetic or higher-income individuals, while impulsive cues remain effective for highly empathetic donors. The findings can inform the design of food bank fundraising strategies that increase charitable giving and strengthen food security during crises.

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Keywords

Information nudges; deliberate giving; impulsive giving; COVID-19; charity donation; food banks; intertemporal giving choices; food security

JEL classification

C91; D64; D12

Declarations of interest: None

1. Introduction

The application of nudges in charity donations has emerged as a strategic approach to influence and enhance philanthropic behaviors (Thaler & Sunstein, 2009). Prior research shows that well-designed nudges can increase both the frequency and the amount of charitable donations (Bhargava & Loewenstein 2015). Nudges can be classified into two types based on the decision-making processes they target: deliberate and impulsive nudging. Deliberate nudging focuses on individuals' reflective and analytical thinking processes. It may provide clearer information, restructure complex choices, or offer feedback, thereby facilitating more informed and thoughtful decisions (Sunstein, 2016; Hansen, 2016; Karlan et al., 2019; Schulz et al., 2018). In contrast, impulsive nudges interact with people's automatic and intuitive decision-making systems (Thaler & Sunstein, 2021). These nudges evoke immediate emotional or instinctual reactions that bypass deliberate contemplation (Västfjäll et al., 2014; Rook & Fisher, 1995; Strack & Deutsch, 2004). In this study, we designed an incentive-compatible experiment employing two distinct formats of information: a textual paragraph requiring analytical processing of the contribution's value (deliberate nudge) and a photograph depicting a food bank queue (impulsive nudge). We also evaluated the impact of donation timing, i.e., when the donation takes place, on donation rates. Understanding the intricacies of intertemporal decision-making can help charitable organizations design nudges that either capitalize on immediate generosity or foster sustained future commitments.

Exploring both deliberate and impulsive nudges is important, as they engage distinct cognitive pathways and may differ in effectiveness depending on context and audience. By understanding how these nudges operate, charitable organizations can design more holistic and effective fundraising strategies (Sunstein, 2016). While providing donors with detailed information about the impact of their donations can encourage more substantial, thought-out contributions (Bhargava & Loewenstein, 2015), an emotionally charged image or story in a donation appeal can often elicit an immediate empathetic response and desire to help (Small & Simonsohn, 2007; Bekkers & Wiepking, 2011; Taute & McQuitty, 2004). Only a few studies have examined the relative effects of deliberation and impulsivity on donation decisions, yielding mixed conclusions. Adena and Hager (2020) found that donations to a children's charity were larger in response to an informative (deliberate) video advertisement than to an emotional (impulsive) one. In contrast, Bennett (2009) showed that a hospice organization received higher donations when its website used emotionally oriented imagery rather than informational content.

Intertemporal decision-making refers to the process by which individuals evaluate outcomes occurring at different points in time, typically involving trade-offs between immediate and future rewards or costs. Such decisions are influenced by temporal discounting, where individuals tend to devalue future outcomes in favor of immediate ones (Frederick et al., 2002). When applied to charitable giving, intertemporal decisions often involve choosing whether to donate now or later. Organizations might use nudges to emphasize the immediate impact of donating, making present-day benefits more salient and spurring immediate contributions (Thaler & Sunstein, 2021). Alternatively, charities can employ commitment devices or offer incentives to encourage donors to pledge future donations, appealing to the desire to make a sustained impact (Rogers & Bazerman,

2008). Many studies have shown that donations increase significantly when donors are asked to give in the future (Andreoni & Sera-Garcia, 2021; Breman, 2011; Sparrow & Spaniol, 2018). Conversely, others find evidence of present bias, indicating that people often prefer immediate giving over delayed commitments (Ariely & Zauberman, 2003).

Previous research has identified several personal characteristics associated with charitable giving behavior (Adena et al., 2019; Adena & Huck, 2019; Bekkers & Wiepking, 2011; Drouvelis & Marx, 2018, 2021; Schlegelmilch et al., 1997; Huck et al., 2015; Lee & Chang, 2008; List, 2011; Marx, 2018, 2021; Small & Simonsohn, 2008). Among these, empathy, defined as compassion or concern for others, is one of the most studied traits. Empathy has consistently been found to positively influence charitable giving (De Waal, 2012; Kamas & Preston, 2021; Lee et al., 2014; Li et al., 2019; Liu et al., 2018; Kim & Kou, 2014). For example, Andreoni et al. (2018) found that a one standard deviation increase in empathetic concern was associated with an 8% increase in the likelihood of giving. Moreover, Bickel et al. (1999) found that personal impulsiveness is positively associated with the time discount rate. Individuals' time discount rate and risk attitude were often found to affect their intertemporal choices (e.g. Andersen et al. 2008; Zhang et al. 2019). Therefore, in our study, we examine how these personal characteristics, in addition to demographic factors, may influence donation decisions.

Our experiment was conducted during the COVID-19 pandemic, when economic disruptions dramatically increased food insecurity across the United States (Pew Research Center, 2020). The U.S. household food insecurity rate stood at 10.5% (13.7 million households) in 2019 (Coleman-Jensen et al., 2020; Lee et al., 2021), one of the lowest in two decades, but surged to 44% in March 2020 (Wolfson & Leung, 2020). Feeding America, the largest network of food banks in the U.S., reported a 50% rise in demand during the pandemic (Feeding America, 2020, 2021). The COVID-19 crisis (2020–2021) caused food insecurity on a large scale and increased reliance on charitable food organizations (Niles et al., 2020; Fleischhacker & Bleich, 2021). Similar patterns have occurred during other shocks, including the 2018–2019 federal government shutdown, which disrupted SNAP benefit timing and created spillovers into emergency food networks (Gosliner et al., 2020), and cuts in SNAP benefit levels or purchasing power, such as the 2013 reduction associated with the ARRA sunset (Kim et al., 2020). Similar dynamics also appear in more recent disruptions, including the March 2023 expiration of SNAP emergency allotments, which was followed by increased food insufficiency and greater food pantry use (Wells et al., 2024), and reported 2025 interruptions affecting food-bank supply linked to pauses or cancellations in USDA-supported deliveries and related funding streams (Politico, 2025; Reuters, 2025). Taken together, these episodes suggest that disruptions in federal nutrition assistance can quickly shift demand toward charitable food sources and donations, whether triggered by a public health emergency, an administrative breakdown, or policy changes (GAO, 2024; Gray, 2019; Coleman-Jensen et al., 2020). Understanding how individuals make donation decisions and how to motivate more giving is therefore central to ensuring the financial sustainability of food banks under crisis conditions.

This paper makes three key contributions. First, it is among the first to experimentally compare deliberate and impulsive nudging within the context of food bank donations, providing novel evidence on how distinct cognitive mechanisms shape prosocial behavior. Second, by incorporating donation timing (immediate versus delayed) within a within-subject, incentive-compatible design, it offers new insights into intertemporal decision-making in charitable giving, an area with limited empirical evidence. Third, by linking donation behavior to individual characteristics such as empathy, income, and time preferences, this study identifies heterogeneous responses to different nudging strategies. The findings contribute to behavioral and food policy discussions on how to design effective, evidence-based campaigns to enhance food security and support charitable organizations during crises like the COVID-19 pandemic.

The remainder of this paper is organized as follows. Section 2 describes the experimental methodology. Section 3 presents the data and descriptive statistics. Section 4 introduces the econometric models. Section 5 discusses the main results, and Section 6 concludes.

2. Research methodology

We conducted an online donation experiment through Amazon MTurk with 537 participants from across the United States during October 2020, in the midst of the COVID-19 pandemic. The study was approved by the Institutional Review Board of our institution. The sample size was reduced to 527 after excluding the participants who failed the attention checks of simple calculation questions. A total of 262 participants were randomly assigned to the deliberate nudging treatment group, while the other 265 participants were in the impulsive nudging treatment group. The number of participants in our study is comparable to the sample size in similar studies. For example, in Andreoni & Serra-Garcia (2021), 31% of 179 participants in the Donate Now group and 45% of 173 participants in the Donate Later group chose to donate their participation payment. The estimated sample size from the two-sided power calculation based on Andreoni & Serra-Garcia (2021) with a power of 0.8 is 188 subjects per treatment group (149 per treatment for one-sided). In Adena et al. (2019), the total number of participants was 486 with 245 participants in the control group and 241 participants in the treatment group. In Zarghamee et al. (2017), a total of 187 students participated in a donation study where 69 were in the Donation treatment and 118 were in the Refund treatment.

The participants earned a \$2 base payment for completing the survey and were also given an extra \$1 endowment with which they could choose either to donate this extra money to food banks or to keep it for themselves. The participants were randomly assigned to either a deliberate or impulsive treatment group. Our study is designed to compare two decision environments rather than estimate the absolute effect of nudging relative to no intervention. Both conditions represent realistic behavioral contexts, so the primary objective was to identify which type of nudge is more effective at influencing donation behavior rather than whether nudging itself works.

At the beginning of the survey, all participants from both groups were presented with the same information regarding food insecurity during the COVID-19 pandemic, “According to Feeding

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America, a nationwide network of more than 200 food banks and 60,000 food pantries, more than 54 million people may experience food insecurity in 2020 due to the effects of the coronavirus pandemic.” Thereafter, as shown in Figure 1, the deliberate and impulsive treatment groups received different information. The deliberate treatment group were shown analytic information quantifying the contribution’s value, stating that every one dollar donated provides ten meals, as reported by Feeding America on its donation page. The impulsive treatment group were shown a photo of cars lined up and packed tightly in a food bank parking lot. We used a photo of cars rather than one depicting people to avoid potential confounds associated with visual cues of identity (Winterich et al., 2009).

Deliberate treatment group	Impulsive treatment group
Your impact: \$1 = 10 meals Every dollar you give can provide at least 10 meals to families in need through the Feeding America network of food banks	Below is a photo showing a sea of cars lined up for food assistance from a local food bank during the COVID-19 outbreak.

Fig 1: Nudging information by treatment group

In addition to the between-subject design with nudging information, a within-subject design was employed to examine intertemporal donation decisions (Andreoni & Serra-Garcia, 2021), as shown in Figure 2. After viewing either the deliberate or impulsive information nudge according to their assigned treatment group, participants were presented with two questions that asked about their willingness to donate either *in a month* or *on the same day* (displayed as “today” on the experiment screen). These two questions appeared on separate screens in a random sequence. At the end of the survey, one of the two responses was randomly selected as the binding decision, and the corresponding donation or payment to the participant was implemented accordingly. To assure participants that the donations would be carried out as described, we provided a website address where the donation receipts were publicly posted.

Donate on the same day	Donate in a month
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If you are receiving \$1 today (this is in addition to the \$2 participation fee), would you like to donate this \$1 to Food Banks through us? ☐ NO, I do not want to donate. I would like to keep the given \$1. ☐ YES, I would like to donate the given \$1.	If you are receiving \$1 in one month (this is in addition to the \$2 participation fee), would you like to donate this \$1 to Food Banks through us? ☐ NO, I do not want to donate. I would like to keep the given \$1. ☐ YES, I would like to donate the given \$1.
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Fig. 2. Intertemporal donation decisions

To test whether the amount of the endowment affected participants' decisions (e.g., Depositario et al., 2014), 65 participants were randomly selected to receive an extra \$2 instead of \$1. We found that the amount of the endowment did not significantly affect participants' decisions. The main hypotheses of our study are as follows:

Hypothesis 1: Deliberate and impulsive types of information have different effects on giving rates among different types of people.

Hypothesis 2: People are more likely to donate in a month than to donate today.

We also collected demographic information and personal characteristics. Time discount factors and risk preferences were measured following previous studies (e.g., Andersen et al., 2008; Dave et al., 2010; Eckel & Grossman, 2008), and the specific values used in the questionnaire were adjusted to the context of this study. Empathy was measured following Schlegelmilch et al. (1997) and Lee and Chang (2008). Details of how we measured time discount rate, risk preference, food assistance experience, and empathy are provided in the Appendix. Participants' trust in the experiment's commitment to donate the specified amount to food banks was measured by their level of agreement with the statement, "We will donate the amount you indicated above to food banks."

3. Summary statistics

Table 1 presents the summary statistics. Participants were randomly assigned to either the impulsive treatment group or the deliberate treatment group. Overall, the sample was well balanced in observable characteristics across the two groups. About 66% of participants in both groups were classified as having "high empathy" based on their responses to question A4 in the Appendix. The

difference in the proportions of high-empathy individuals across groups was not statistically significant ($p = 0.998$). The proportions of participants who reported having previously received food assistance were 54.6% and 54.3% for the deliberate and impulsive groups, respectively ($p = 0.956$). In addition, 73.3% of participants in the deliberate group and 72.8% in the impulsive group agreed (somewhat or strongly) with the statement “Donations are a way to reduce social or economic inequality” ($p = 0.907$). About 52.3% of participants in the deliberate group and 47.2% in the impulsive group were classified as “not risk-averse” according to their answers to question A2 in the Appendix ($p = 0.241$). The proportions of those who trusted the experimenter’s donation commitment—agreeing with the statement “We will donate the amount you indicated above to food banks.”—were 85.9% in the deliberate group and 81.1% in the impulsive group ($p = 0.143$).

The demographic characteristics were also generally well balanced across the treatment groups. Females accounted for 38.9% of the deliberate group and 35.8% of the impulsive group ($p = 0.456$). Similarly, 71.0% and 70.6% of participants were married in the deliberate and impulsive groups, respectively ($p = 0.915$). We divided income into two categories: lower income ($\leq \$50,000$) and middle/upper income ($> \$50,000$) (Snider, 2000). About 43% of participants were in the lower-income group, with the remainder in the middle/upper-income group. Income distributions were well balanced across the two treatment groups. At the time of the survey (October 2020), the employment rates were 92.7% and 93.6% for the deliberate and impulsive groups, respectively ($p = 0.704$). Racial composition was also similar across groups: 72.9% of participants in the deliberate group and 73.6% in the impulsive group identified as White, with no statistically significant difference ($p = 0.860$). The average number of children per household was 2.095 in the deliberate group and 2.072 in the impulsive group ($p = 0.804$). Religion was coded as a binary variable indicating whether an individual attended religious services at least once a week. The proportion of regular attendees was 46.6% in the deliberate group and 40.0% in the impulsive group ($p = 0.129$). Political standing was also measured as a binary variable (1 = conservative, 0 = otherwise) and was nearly identical across groups: 55.0% and 54.7% for the deliberate and impulsive groups, respectively ($p = 0.955$). However, the COVID-19 infection rate differed between the two groups, with 17.6% of participants in the deliberate group and 24.9% in the impulsive group reporting prior infection ($p = 0.039$).

In our survey, we asked participants about their household food security status. The results indicate that food security was significantly threatened during the COVID-19 pandemic (Figure 3). Approximately 21% of those who had reported having “enough and preferred” food prior to the pandemic reported having “enough but not preferred” or “sometimes not enough” food during the pandemic. The proportion of food-secure participants who had “enough and preferred” food decreased from 59.01% (311 out of 527 participants) before the pandemic to 45.73% (241 participants) afterward. In contrast, the proportion of participants who reported having “enough but not always their preferred” food increased from 25.8% (136 participants) before COVID-19 to 37.38% (197 participants) during the pandemic. The share of participants who reported not having enough food also rose, from 12.14% before the pandemic to 14.23% during COVID-19.

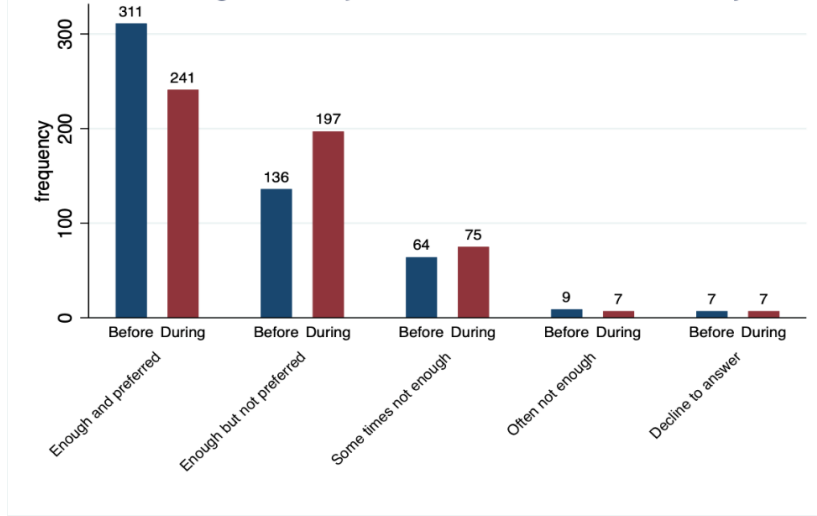


Fig. 3. Household food security before and during the COVID-19

4. Model specification

As mentioned above, we asked participants two randomly ordered willingness to donate questions (i.e., in a month or the same day). Hence, we used Seemingly Unrelated Bivariate Probit regression model (SUR) to identify important factors that affect donation decisions. By treating 'donate *same day*' and 'donate *in a month*' as two separate dependent variables, we constructed a system of two Probit regressions with identical control variables as below.

$$\begin{bmatrix} Y_1 \\ Y_2 \end{bmatrix} = \begin{bmatrix} X_1 & 0 \\ 0 & X_2 \end{bmatrix} \begin{bmatrix} \beta_1 \\ \beta_2 \end{bmatrix} + \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

$$\beta^{SUR} = [X' \Omega^{-1} X]^{-1} [X' \Omega^{-1} Y]$$

where Y_j is a binary variable that indicates donation decisions of the participants, and $j = \{1, 2\}$ represents the 'same day' and 'in a month' decisions respectively, and X_j is a vector of explanatory variables. Since each participant answered the two questions (donate on the same day and donate in a month), SUR model is appropriate in this intertemporal research design because the error terms are highly likely to be correlated across equations within individual and likely to be uncorrelated across individuals. The correlation coefficient between residuals of the two Probit models were significantly different from 0, which statistically supports the use of SUR bivariate Probit model. Here, u_{ij} is assumed to have zero mean, homoskedastic, and independent across individual choices. For a given individual, the covariance of the error term across equations is:

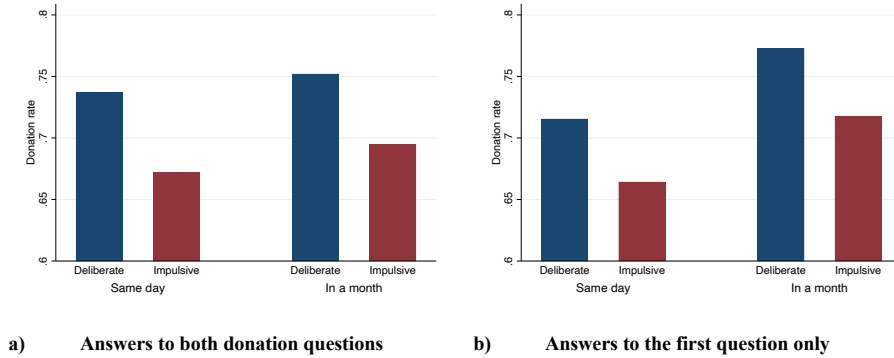
$E[u_{ij}u_{ij'}|X] = \sigma_{jj'} * I_N \neq 0$ where $j \neq j'$. Ω^{-1} is a weighting matrix based on the covariance of the error terms.

Probit regressions were applied for subgroup analysis to evaluate heterogeneous effects of information treatments. SUR were not applied due to smaller sample size in each subgroup. The Probit regressions included all variables used in the SUR models plus the ‘donation time’ dummy variable (i.e., 0 if ‘donation *same day*’; 1 if ‘donation *in a month*’), and an interaction term between information treatment and donation time.

5. Results

5.1 Donation rates by treatments

We analyzed the data by calculating the donation rate, defined as the percentage of respondents who chose to donate in each of the two willingness-to-donate questions (“Would you like to donate today?” and “Would you like to donate in a month?”). As shown in Figure 4, participants were generally more likely to donate when the donation was scheduled for a month later than when it was immediate, and donation rates were higher under the deliberate nudging condition.



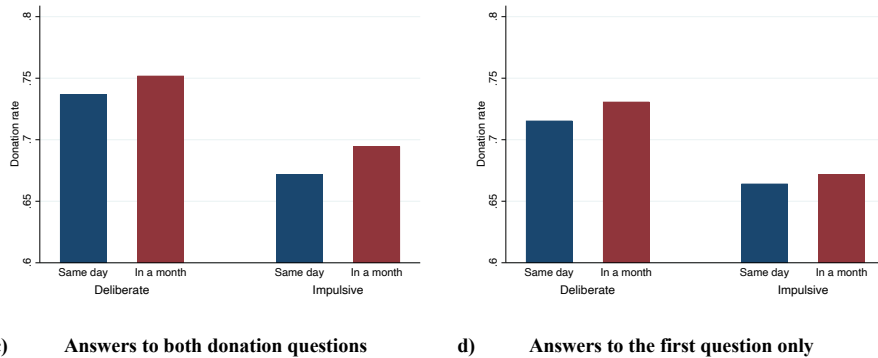


Fig. 4. Donation rate by information treatment and donation time

The donation rate among participants who received deliberate information was significantly higher than among those who received impulsive information (74.4% vs. 68.3%, $p = 0.028$). Participants were also more likely to donate when the donation was to be made in a month rather than on the same day (72.3% vs. 70.4%, $p = 0.070$).

Because the two donation questions were randomly ordered, the first question might have influenced responses to the second. To test for this potential order effect, we re-estimated donation rates using only participants' answers to the first question. The results were consistent: the deliberate information group still showed a higher donation rate than the impulsive group (74.4% vs. 69.1%, $p = 0.086$), and the rate for donations made in a month remained higher than for same-day donations (74.5% vs. 68.9%, $p = 0.078$).

5.2 Heterogeneity of treatment effects across subgroups

Figure 5 presents the heterogeneity of treatment effects across subgroups with different personal characteristics, pooling responses to both the same-day and one-month donation questions.

5.2.1 Empathy

Figure 5(a) illustrates that although the overall donation rate was higher among participants with high empathy, the increase associated with deliberate nudging was greater for those with low empathy. For individuals with low empathy, the difference in donation rates between the deliberate and impulsive treatment groups was statistically significant ($p = 0.026$), whereas for those with high empathy, the difference was not significant. One possible explanation is that individuals with high empathy already have a strong intrinsic motivation to donate, and so as a result, deliberate nudging has limited additional influence on their behavior.

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5.2.2 Experience in receiving food assistance

Figure 5(b) shows that the donation rate among participants who had previously received food assistance was substantially higher (85.7%) than that among those who had not (54.2%) ($p = 0.000$). This finding suggests that individuals who had received food assistance were not only beneficiaries but also more likely to act as benefactors. Among participants without prior experience in food assistance programs, the donation rate under deliberate nudging (60.1%) was significantly higher ($p = 0.005$) than under impulsive nudging (48.3%), whereas the treatment effect was not significant for those with prior food assistance experience. It is possible that individuals who previously received food assistance may have developed stronger norms of reciprocity, gratitude, or empathy, motivating them to “pay it forward”. In contrast, individuals without such experience may rely more on deliberative reflection about social responsibility, making them more responsive to deliberate nudging.

5.2.3 COVID-19 infection experience

Figure 5(c) indicates that participants with a history of COVID-19 infection exhibited higher donation rates (90.6%) than those without such history (66.1%) ($p = 0.000$). For participants who had not been infected previously, the donation rates were 70.8% under deliberate nudging and 61.1% under impulsive nudging, indicating that deliberate nudging was more effective in promoting donations ($p = 0.002$). Among those who had been infected, donation rates were high in both treatment groups (91.3% and 90.2% for the deliberate and impulsive groups, respectively; $p = 0.386$), and the treatment effect was not significant. A possible reason for the elevated donation rates among participants with infection history is that personal exposure may have heightened their awareness of the pandemic's social and economic impacts. These findings align with previous literature: Adena and Harke (2022) found that donation rates were higher in areas where the pandemic was severe and widely covered by the media, and Grimalda et al. (2021) reported that individuals with personal exposure to COVID-19 were more likely to donate than others.

5.2.4 Risk preference

As illustrated in Figure 5(d), risk-averse individuals were more inclined to donate compared to others, with donation rates of 76.6% and 66.0%, respectively ($p = 0.000$). Notably, deliberate nudging was more effective than impulsive nudging among those who were not risk averse, with donation rates of 71.5% and 60.0%, respectively ($p = 0.005$). In contrast, among risk-averse participants, the difference between treatments was not statistically significant, although the donation rate was slightly higher in the deliberate group. Risk-averse individuals may be more inclined to donate because giving can reduce feelings of moral or social risk, helping maintain a positive self-image or avoid potential regret from not helping. In contrast, individuals who are less risk-averse may rely more on immediate preferences, so prompts that encourage deliberation can increase donations by prompting reflection on social norms or the benefits of helping others.

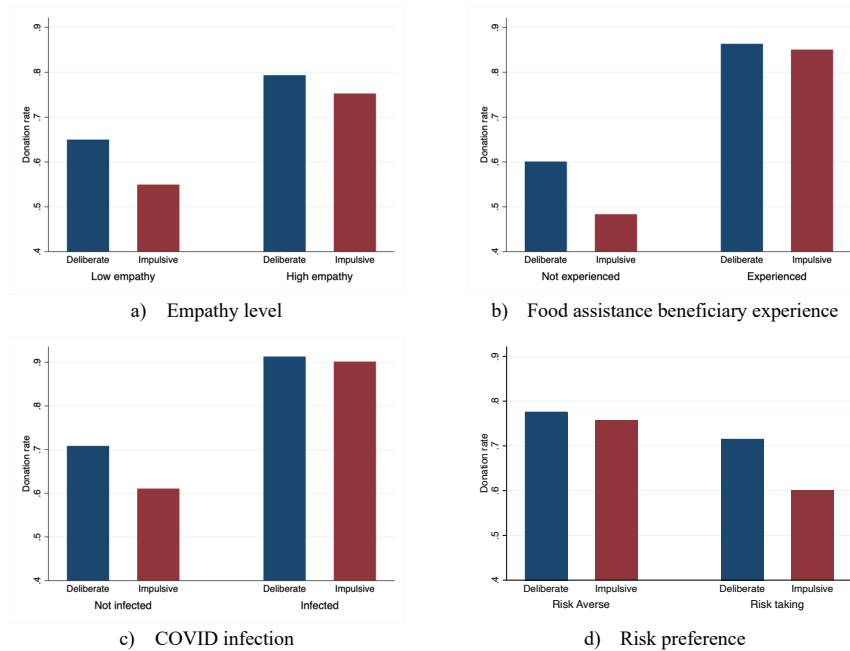
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5.2.4 Income

We classified participants into two income groups: a lower-income group ($\leq \$50,000$) and a middle/upper-income group ($> \$50,000$) following Snider (2000) (Figure 5e). Among middle- and upper-income participants, the donation rate under deliberate information (81.1%) was significantly higher than under impulsive information ($p = 0.002$). Among lower-income participants, the donation rate was slightly higher in the deliberate group but the difference was not statistically significant ($p = 0.396$).

5.2.5 Belief that donations reduce inequality

Figure 5(f) shows that the donation rate was higher among participants who believed that donations help reduce social or economic inequality. Furthermore, among those who did not share this belief, deliberate nudging resulted in significantly higher donations than impulsive nudging (63.6% vs. 52.8%, $p = 0.033$).



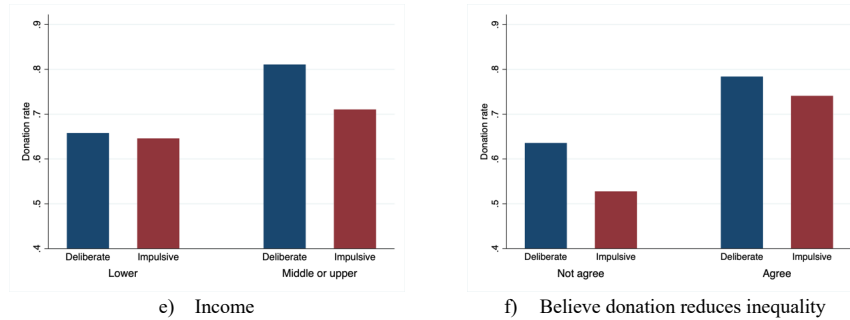


Fig. 5. Heterogeneity in Information Treatment Effects

5.3 Seemingly Unrelated Probit Regression

We further conducted Seemingly Unrelated Probit Regression (SUR) to evaluate the treatment effects and important factors that affect donation rates controlling for variables such as age, gender, income, marital status, race, number of children, employment, political ideology, religion, time discount rate, and risk preference. As a robustness check, we also conducted analysis using random effect Probit regression modeling donation decisions in different times as correlated responses, and it yielded similar results to the SUR model. Additionally, we conducted additional robustness check by only using participants' answers to the first intertemporal donation question they received, and similar results were found. We chose to present SUR results since the values of the correlation coefficients ('athrho') for both groups are significant.

The SUR results (Table 2) showed that when pooling all samples together, deliberate nudging is more effective than impulsive nudging in promoting donation, although this effect is statistically significant on the same day but not in a month. Participants with lower time discount were less likely to give than others when asked to donate on the same day. This effect is not significant when the donation was to be made in a month. This result is consistent with the fact that people with lower time discount rate have higher propensity to save for the future (Andreoni & Serra-Garcia, 2021; Breman, 2011; Frederick et al., 2002).

Empathy level is positively related with donation rates, consistent with previous literature (Andreoni et al., 2018; Kim and Kou, 2014; Li et al., 2019; Liu et al., 2018; Verhaert and Van den Poel, 2011). The overall magnitude of the positive association was greater for the impulsive group than the deliberate group for both donation on the same day and donation in a month. The SUR results show that high empathy level was associated with a 14.7% ($p = 0.003$) higher donation rate with impulsive information and 10.7% ($p = 0.050$) higher donation rate with deliberate information compared to low empathy level. Donation rates did not differ between participants who received \$1 and those who received \$2. We also found that experience in receiving food

assistance had a significant positive association with the donation rate in all regressions. Those who received food assistance were 17.3% more likely to donate ($p = 0.000$), implying that many beneficiaries were benefactors as well. Additionally, people who are married (17.0%, $p = 0.000$), more conservative (11.2%, $p = 0.001$), or had been infected by COVID (15.1%, $p = 0.004$), were more likely to donate than others. Interestingly, people with higher income were more likely to donate on the same day with deliberate nudging but not with impulsive nudging, or when donation was to be made in a month.

5.4 Subgroup analysis

To examine how the effects of nudging varied across individual characteristics, we conducted subgroup analyses using Probit models (Table 3). A robustness check based on participants' responses to just the first intertemporal donation question produced consistent results. Among participants with low empathy, deliberate nudging increased the donation rate by 10.3% compared to impulsive nudging ($p = 0.091$), although this difference was not significant for highly empathetic individuals. For participants who had never received food assistance, deliberate nudging raised the donation rate by 11.0% ($p = 0.052$), while no significant difference was observed among those who had previously received such assistance ($p = 0.753$). Among participants who trusted that the experimenter would donate to food banks as promised, deliberate nudging increased donations by 8.4% relative to impulsive nudging ($p = 0.017$). In contrast, among those who did not express trust, impulsive nudging produced an 11.1% higher donation rate, although the effect was not statistically significant ($p = 0.207$).

For participants without a history of COVID-19 infection, deliberate nudging raised the donation rate by 7.8% ($p = 0.050$), whereas no significant effect was found among those who had been infected. Across income groups, deliberate nudging was more effective than impulsive nudging among middle- and upper-income participants (10.3% higher, $p = 0.013$), but not among lower-income participants. Finally, deliberate nudging was also more effective among those who did not believe that donations reduce inequality (14% higher, $p = 0.049$), whereas no difference was observed among those who held that belief.

[Because Table 3 presents treatment effects across multiple subgroup specifications, we apply the Benjamini–Hochberg procedure to control the false discovery rate across the 12 p-values associated with the Treatment row. When controlling the false discovery rate at the 10 percent level, the treatment effect remains statistically significant for two subgroups: respondents in the Trust: Agree group and those in the Income: Middle/Upper group, both with adjusted q-values of 0.099. The remaining subgroup estimates are not statistically significant after the multiple-testing correction. Nevertheless, the direction and magnitude of the subgroup estimates remain broadly consistent with the interpretation that treatment effects are stronger among respondents with higher levels of trust and relatively higher income.](#)

6 Conclusion

This study provides experimental evidence on how different types of informational nudges and donation timing influence charitable giving during a period of heightened food insecurity. Using an incentive-compatible design, we found that deliberate nudging, which provides analytical information about the impact of donations, was generally more effective than impulsive nudging based on emotional imagery. Donation rates were also higher when individuals were asked to donate in the future rather than immediately, suggesting that allowing time for reflection or financial planning can enhance giving intentions. Importantly, the effectiveness of nudges varied across individual characteristics, demonstrating heterogeneous treatment effects. Deliberate nudging was particularly effective among individuals with lower empathy, higher income, or greater trust in the donation process, while impulsive nudging had a stronger influence on those with higher empathy. Individuals with lower time-discount rates were less likely to donate immediately but equally likely to give when the donation was delayed. Moreover, beneficiaries of food assistance programs and those affected by COVID-19 were significantly more likely to donate, suggesting that personal experience with hardship strengthens prosocial motivation.

These results provide insights into how behavioral interventions can be tailored to enhance charitable giving. By accounting for individual differences in response to nudges and timing, policymakers and nonprofit organizations can design evidence-based, context-sensitive strategies that strengthen food security systems and foster resilient community support during times of crisis. Introducing “donate later” or scheduled giving options could sustain engagement, particularly during economic uncertainty. Policymakers and nonprofit managers can also design behaviorally informed campaigns that combine cognitive and emotional appeals while recognizing the role of trust and prior beneficiary experience in shaping giving behavior.

Food banks often serve as a primary buffer when households experience sudden increases in food hardship. During major disruptions such as the COVID-19 pandemic, federal government shutdowns, and periods of reduced SNAP support, demand for emergency food assistance can rise quickly, while staffing, inventories, and funding do not expand at the same pace (Fleischhacker & Bleich, 2021; Gray, 2019; USDA, 2019; Larson & McLachlin, 2011), shift a larger share of assistance responsibilities to charitable providers. Food banks need ways to stabilize resources under volatile conditions, especially when funding and donation flows are uncertain. Our study addresses this need by testing how nudging type and donation timing shape giving. The results identify strategies that increase participation and improve the predictability of support during crises, reducing vulnerability to abrupt swings in food security.

This study has some limitations that should be acknowledged. While we examined the difference between deliberate nudging and impulsive nudging, we did not compare the effectiveness of these nudges to a control condition where no nudge is provided. This design choice reflects the study’s primary objective of comparing the relative effectiveness of two commonly observed decision environments: one that encourages deliberation and one that relies on more spontaneous decision-making, rather than estimating the absolute effect of nudging compared to no intervention. Nevertheless, future research should test the robustness of our findings by

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incorporating a control condition, and also investigating the cognitive processes that subjects employ when processing nudging information. ▼

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Table 1
Summary statistics

Variable	Mean			Balance Check
	Pooled	Deliberate	Impulsive	P-value
Treatment 0=Impulsive group, 1=Deliberate group	0.497			
Endowment amount 1= \$1, 2= \$2	1.123	1.126	1.121	0.856
Donation decision same day & in a month 1= Donate; 0= Not to Donate	0.713	0.744	0.683	0.028
Donation decision same day 1= Donate; 0= Not to Donate	0.704	0.737	0.672	0.103
Donation decision in a month 1= Donate; 0= Not to Donate	0.723	0.752	0.694	0.14
Empathy 0=Low empathy, 1=High empathy	0.657	0.656	0.657	0.998
Experience in food assistance 1= Experienced in receiving food assistance, 0=Otherwise	0.545	0.546	0.543	0.956
Believe that donation reduces inequality 0=Not agree, 1=Agree	0.731	0.733	0.728	0.907
Time discount 1=Low time discount, 0=Otherwise	0.165	0.179	0.151	0.380
Risk preference 0=Not Risk-averse, 1=Risk-averse	0.497	0.523	0.472	0.241
Trust in donation commitment 1=Trust that the experimenter will donate as promised, 0=otherwise	0.835	0.859	0.812	0.143
Gender 0=Male, 1=Female	0.374	0.389	0.358	0.456
Marital status 1=Married, 0=Otherwise	0.708	0.71	0.706	0.915
Income (in \$1,000) Household annual income in 2019	60.133	59.408	60.849	0.603
Age	37.438	38.344	36.543	0.058
Employment status 1=Employed, self-employed, 0=Otherwise	0.932	0.927	0.936	0.704
COVID-19 infection history 1=Infected at the time of survey or before, 0=Otherwise	0.213	0.176	0.249	0.039
Race 1=White, 0=Otherwise	0.732	0.729	0.736	0.86
Number of children in household	2.083	2.095	2.072	0.804
Religion 1= Attend religious service once a week or more, 0=Otherwise	0.433	0.466	0.4	0.129
Political standing 1=Conservative, 0=Otherwise	0.548	0.55	0.547	0.955

Table 2

Seemingly Unrelated Probit Regression Models of Donation Rates

	Regression (1) Pooled			Regression (2) Deliberate			Regression (3) Impulsive		
	Coef. Same day	Coef. In a month	Margins	Coef. Same day	Coef. In a month	Margins	Coef. Same day	Coef. In a month	Margins
Treatment	0.226 [*]	0.215	0.061 [*]						
0=Impulsive, 1=Deliberate	(0.132)	(0.133)	(0.034)						
Empathy level	0.434 ^{***}	0.447 ^{***}	0.122 ^{***}	0.303	0.494 ^{**}	0.107 [*]	0.544 ^{***}	0.579 ^{***}	0.147 ^{***}
0=Low, 1=High	(0.146)	(0.146)	(0.037)	(0.217)	(0.218)	(0.054)	(0.208)	(0.210)	(0.050)
Believe donation reduce inequality	0.209	0.307 ^{**}	0.070 [*]	0.026	0.339	0.046	0.381 [*]	0.239	0.086 [*]
0=Not agree, 1=Agree	(0.148)	(0.149)	(0.038)	(0.226)	(0.226)	(0.058)	(0.209)	(0.210)	(0.051)
Food assistance experience	0.642 ^{***}	0.602 ^{***}	0.173 ^{***}	0.670 ^{***}	0.548 ^{**}	0.169 ^{***}	0.601 ^{***}	0.640 ^{***}	0.163 ^{***}
1= Experienced in receiving food assistance, 0=Otherwise	(0.150)	(0.152)	(0.037)	(0.217)	(0.215)	(0.053)	(0.220)	(0.223)	(0.052)
Time discount	-0.315 [*]	-0.187	-0.072	-0.232	0.033	-0.031	-0.251	-0.451	-0.087
1=Low time discount, 0=Otherwise	(0.176)	(0.179)	(0.045)	(0.246)	(0.260)	(0.064)	(0.278)	(0.281)	(0.068)
Risk preference	-0.035	0.072	0.003	0.043	0.312	0.045	-0.071	-0.180	-0.030
0=Not Averse, 1=Averse	(0.140)	(0.140)	(0.036)	(0.201)	(0.203)	(0.052)	(0.201)	(0.201)	(0.050)
Trust in donation commitment	0.182 ^{***}	0.093	0.040 ^{***}	0.203 ^{**}	0.065	0.039 [*]	0.160 [*]	0.107	0.037 [*]
0=Strongly disagree...4=Strongly agree	(0.060)	(0.061)	(0.015)	(0.089)	(0.091)	(0.023)	(0.083)	(0.084)	(0.020)
Gender	0.137	0.090	0.032	0.143	-0.095	0.010	0.083	0.267	0.041
0=Male, 1=Female	(0.140)	(0.140)	(0.036)	(0.207)	(0.201)	(0.052)	(0.200)	(0.204)	(0.049)
Marital status	0.590 ^{***}	0.643 ^{***}	0.170 ^{***}	0.600 ^{**}	0.817 ^{***}	0.191 ^{***}	0.688 ^{***}	0.549 ^{**}	0.167 ^{***}
1=Married, 0=Otherwise	(0.173)	(0.174)	(0.043)	(0.251)	(0.258)	(0.063)	(0.246)	(0.251)	(0.058)
Income	0.001	-0.001	0.000	0.007 ^{**}	0.001	0.001	-0.003	-0.002	-0.001
In \$1,000	(0.002)	(0.002)	(0.001)	(0.003)	(0.003)	(0.001)	(0.003)	(0.003)	(0.001)
Age	-0.005	-0.003	-0.001	-0.002	-0.003	-0.001	-0.009	-0.005	-0.002
	(0.006)	(0.006)	(0.002)	(0.009)	(0.009)	(0.002)	(0.009)	(0.009)	(0.002)
Employment	0.098	0.007	0.016	-0.038	-0.133	-0.022	0.214	0.356	0.071
1=Employed, 0=Otherwise	(0.249)	(0.253)	(0.064)	(0.357)	(0.356)	(0.092)	(0.377)	(0.378)	(0.092)
COVID infection	0.441 ^{**}	0.684 ^{***}	0.151 ^{***}	0.360	0.848 ^{**}	0.159 [*]	0.396	0.589 ^{**}	0.125 [*]
1=Ever infected, 0=Otherwise	(0.200)	(0.216)	(0.052)	(0.301)	(0.345)	(0.082)	(0.278)	(0.289)	(0.069)
Race	0.063	0.025	0.013	0.071	0.244	0.041	-0.142	-0.113	-0.034
0=Non-white, 1=White	(0.157)	(0.161)	(0.041)	(0.241)	(0.242)	(0.061)	(0.232)	(0.234)	(0.057)
Children	-0.026	-0.010	-0.005	-0.143	-0.004	-0.022	0.069	-0.020	0.009
Number of children in household	(0.075)	(0.077)	(0.019)	(0.104)	(0.104)	(0.026)	(0.120)	(0.119)	(0.029)
Religion	0.117	0.060	0.026	0.162	0.200	0.049	0.080	-0.072	0.005
1=Religious service once a week or more, 0=Otherwise	(0.145)	(0.147)	(0.038)	(0.201)	(0.202)	(0.052)	(0.221)	(0.227)	(0.055)
Political standing	0.455 ^{***}	0.338 ^{**}	0.112 ^{***}	0.439 ^{**}	0.022	0.069	0.518 ^{**}	0.606 ^{***}	0.146 ^{***}
1=Conservative, 0=Otherwise	(0.140)	(0.141)	(0.035)	(0.203)	(0.205)	(0.053)	(0.204)	(0.207)	(0.048)
Payment amount	0.117	-0.043	0.013	0.211	0.125	0.047	-0.106	-0.213	-0.039
1=\$1, 2=\$2	(0.197)	(0.196)	(0.051)	(0.291)	(0.294)	(0.076)	(0.288)	(0.280)	(0.069)
Constant	-1.710 ^{***}	-1.045 ^{**}		-1.533 ^{**}	-1.016		-1.352 ^{**}	-1.112 [*]	
	(0.476)	(0.478)		(0.729)	(0.730)		(0.660)	(0.655)	
athrho	1.657 ^{***}			2.599 ^{***}			1.816 ^{***}		
	(0.156)			(0.635)			(0.272)		
N	527		527	262		262	265		265

Table 3
Subgroup Analysis: Probit Model

	Empathy		Food assistance experience		Trust in donation commitment	
	Low	High	No	Yes	Not Agree	Agree
Treatment	0.103*	0.043	0.110*	0.011	-0.111	0.084**
0=Impulsive, 1=Deliberate	(0.061)	(0.039)	(0.057)	(0.036)	(0.088)	(0.035)
<i>q-bh</i>	0.183	0.396	0.125	0.868	0.355	0.099
Donation time	0.045	0.013	0.027	0.016	0.105	0.005
0=Same day, 1=In a month	(0.057)	(0.038)	(0.054)	(0.037)	(0.078)	(0.034)
Treatment*Donation time	-0.033	0.003	-0.003	-0.008	-0.002	-0.005
	(0.085)	(0.055)	(0.080)	(0.052)	(0.125)	(0.050)
Empathy level	-	-	0.162***	0.050*	0.023	0.113***
0=Low, 1=High	-	-	(0.043)	(0.030)	(0.076)	(0.027)
Believe donation reduce inequality	0.052	0.104***	0.061	0.095**	0.064	0.073**
0=Not agree, 1=Agree	(0.045)	(0.033)	(0.046)	(0.032)	(0.070)	(0.029)
Food assistance experience	0.173***	0.163***	-	-	0.394***	0.135***
1= Experienced, 0=Otherwise	(0.048)	(0.031)	-	-	(0.066)	(0.028)
Time discount	-0.085	-0.088**	-0.176***	0.009	-0.033	-0.084***
1=Low time discount, 0=Otherwise	(0.056)	(0.035)	(0.055)	(0.033)	(0.084)	(0.031)
Risk preference	-0.093**	0.059*	-0.031	0.049	-0.045	0.005
0=Not Averse, 1=Averse	(0.047)	(0.031)	(0.046)	(0.031)	(0.069)	(0.028)
Trust in donation commitment	0.035**	0.044***	0.055***	0.024**	-	-
0=Strongly disagree...4=Strongly agree	(0.017)	(0.016)	(0.020)	(0.012)	-	-
Gender	0.045	0.021	-0.046	0.043	0.031	0.021
0=Male, 1=Female	(0.047)	(0.029)	(0.043)	(0.028)	(0.068)	(0.026)
Marital status	0.189***	0.145***	0.239***	0.080**	-0.007	0.189***
1=Married, 0=Otherwise	(0.053)	(0.035)	(0.052)	(0.036)	(0.084)	(0.031)
Income	-0.000	0.000	-0.000	0.001	0.000	-0.000
In \$1,000	(0.001)	(0.000)	(0.001)	(0.001)	(0.001)	(0.000)
Age	-0.001	-0.001	0.001	-0.003**	-0.000	-0.001
	(0.002)	(0.001)	(0.002)	(0.001)	(0.004)	(0.001)
Employment	0.003	0.022	-0.069	0.187***	-0.287**	0.023
1=Employed, 0=Otherwise	(0.095)	(0.054)	(0.070)	(0.065)	(0.144)	(0.047)
COVID infection	-0.004	0.267***	0.548**	0.054*	-0.086	0.215***
1=Ever infected, 0=Otherwise	(0.062)	(0.063)	(0.128)	(0.032)	(0.091)	(0.048)
Race	-0.089	0.035	-0.009	0.020	-0.283***	0.074**
0=Non-white, 1=White	(0.059)	(0.033)	(0.056)	(0.031)	(0.068)	(0.031)
Children	-0.016	0.005	-0.010	0.000	-0.006	-0.011
Number of children in household	(0.023)	(0.018)	(0.032)	(0.012)	(0.030)	(0.015)
Religion	0.125**	-0.022	-0.027	0.036	0.088	0.008
1=Religious service once a week or more, 0=Otherwise	(0.050)	(0.032)	(0.054)	(0.027)	(0.069)	(0.029)
Political standing	0.209***	0.059**	0.124***	0.074***	0.182**	0.100***
1=Conservative, 0=Otherwise	(0.044)	(0.029)	(0.043)	(0.028)	(0.072)	(0.026)
Payment amount	0.028	-0.011	0.057	-0.007	0.103	0.016
1=\$1, 2=\$2	(0.070)	(0.041)	(0.064)	(0.036)	(0.112)	(0.037)
N	362	692	480	574	174	880

Note: *q-bh* reports Benjamini–Hochberg adjusted p-values controlling the false discovery rate across the 12 subgroup treatment effects. “Survives BH-FDR (10%)” indicates significance at the 10% false discovery rate level. Average marginal effects; Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

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Table 3 (Cont'd)
Subgroup Analysis: Probit Model

	COVID-19 infection		Income level		Believe donations reduce inequality	
	No	Yes	Lower	Middle/Upper	Do Not Agree	Agree
Treatment	0.078**	-0.005	-0.014	0.103**	0.140**	0.032
0=Impulsive, 1=Deliberate	(0.040)	(0.041)	(0.053)	(0.042)	(0.071)	(0.037)
<i>q-bh</i>	0.05	0.901	0.796	0.013	0.049	0.383
Donation time	0.013	0.051	0.019	0.024	0.058	0.009
0=Same day, 1=In a month	(0.039)	(0.042)	(0.052)	(0.039)	(0.069)	(0.036)
Treatment*Donation time	-0.006	-0.002	0.031	-0.039	-0.100	0.027
	(0.057)	(0.060)	(0.074)	(0.059)	(0.100)	(0.052)
Empathy level	0.084***	0.180***	0.073*	0.085***	0.083	0.116***
0=Low, 1=High	(0.030)	(0.045)	(0.042)	(0.032)	(0.053)	(0.028)
Believe donation reduce inequality	0.112***	-0.080**	0.008	0.109***	-	-
0=Not agree, 1=Agree	(0.032)	(0.040)	(0.042)	(0.033)	-	-
Food assistance experience	0.228***	-0.045	0.088**	0.232***	0.067	0.192***
1= Experienced, 0=Otherwise	(0.031)	(0.051)	(0.043)	(0.033)	(0.059)	(0.030)
Time discount	-0.127***	0.092*	-0.090	-0.067**	-0.096*	-0.061*
1=Low time discount, 0=Otherwise	(0.038)	(0.048)	(0.061)	(0.033)	(0.058)	(0.036)
Risk preference	0.016	-0.048	0.072*	-0.027	0.015	0.001
0=Not Averse, 1=Averse	(0.031)	(0.033)	(0.040)	(0.031)	(0.055)	(0.029)
Trust in donation commitment	0.026*	0.033***	0.048***	0.037***	0.012	0.045***
0=Strongly disagree...4=Strongly agree	(0.014)	(0.011)	(0.018)	(0.014)	(0.020)	(0.013)
Gender	0.039	-0.019	-0.034	0.063**	0.081	0.010
0=Male, 1=Female	(0.029)	(0.030)	(0.041)	(0.030)	(0.055)	(0.027)
Marital status	0.189***	0.079	0.239***	0.092**	0.236***	0.131***
1=Married, 0=Otherwise	(0.036)	(0.049)	(0.049)	(0.040)	(0.058)	(0.036)
Income	0.000	0.000	-	-	-0.001	0.000
In \$1,000	(0.000)	(0.001)	-	-	(0.001)	(0.000)
Age	-0.001	-0.002	0.002	-0.003*	-0.000	-0.001
	(0.001)	(0.002)	(0.002)	(0.001)	(0.003)	(0.001)
Employment	0.013	0.000	-0.003	-0.048	-0.050	0.025
1=Employed, 0=Otherwise	(0.056)	(.)	(0.060)	(0.083)	(0.101)	(0.052)
COVID infection	-	-	0.210***	0.070	0.259***	0.073*
1=Ever infected, 0=Otherwise	-	-	(0.068)	(0.047)	(0.079)	(0.041)
Race	0.066*	-0.054	0.056	0.023	0.058	0.016
0=Non-white, 1=White	(0.036)	(0.044)	(0.049)	(0.035)	(0.072)	(0.032)
Children	-0.019	0.014	0.024	-0.019	-0.022	0.004
Number of children in household	(0.018)	(0.017)	(0.031)	(0.015)	(0.026)	(0.017)
Religion	0.020	-0.005	0.051	-0.009	0.185***	-0.030
1=Religious service once a week or more, 0=Otherwise	(0.033)	(0.036)	(0.044)	(0.035)	(0.059)	(0.030)
Political standing	0.103***	0.085***	0.100**	0.085***	0.067	0.108***
1=Conservative, 0=Otherwise	(0.029)	(0.031)	(0.041)	(0.031)	(0.054)	(0.028)
Payment amount	0.035	-0.049	0.020	-0.034	-0.047	0.014
1=\$1, 2=\$2	(0.044)	(0.041)	(0.064)	(0.041)	(0.074)	(0.040)
N	830	220	454	600	284	770

Note: *q-bh* reports Benjamini-Hochberg adjusted p-values controlling the false discovery rate across the 12 subgroup treatment effects. *Survives BH-FDR (10%)* indicates significance at the 10% false discovery rate level. Average marginal effects; Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

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Deleted: Average marginal effects; Standard errors in parentheses [2]

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Appendix

Questions A1 – A4 measure the time discount rate, risk preference, food assistance recipient experience, and empathy level of the participants.

A1. Time discount rate

Now, **hypothetically**, suppose you have a choice of two payment options: receive a certain amount of money today, or receive a higher amount of money in one year. Please choose your preferred payment option in each row.

	Receive the money today	Receive the money in one year
\$1,000 Today OR \$1,010 in One Year	☐	☐
\$1,000 Today OR \$1,050 in One Year	☐	☐
\$1,000 Today OR \$1,200 in One Year	☐	☐

Note (not shown to participants): In our analysis, a participant is defined as having low time discount if he/she chooses to receive money in one year for all three rows.

A2. Risk preference

Now, **hypothetically**, if you choose to play a gambling game, among Gamble 1 to Gamble 3, mark your most preferred one. (For example, if you choose Gamble 1, then you will earn \$100 for sure. If you choose Gamble 2, then you will earn either \$80 with 50% chance or \$120 with 50% chance.)

- Gamble 1 \$100 100%
- Gamble 2 \$80 50% \$120 50%
- Gamble 3 \$50 50% \$150 50%

Note (not shown to participants): A participant is defined as risk averse if choosing Gamble 1.

A3. Food assistance recipient experience

A person has experiences in food assistance programs if he/she answers “YES” to any of the following questions.

Q. Do you or any member of your household currently receive SNAP or Food Stamp benefits?

Q. Have you or anyone in your household acquired food from Food Banks in the past year **BEFORE the COVID-19 pandemic?**

Q. Have you or anyone in your household acquired food from Food Banks in the past year **DURING the COVID-19 pandemic?**

- ☐ Yes
- ☐ No
- ☐ Not sure
- ☐ Decline to answer

A4. Empathy

The following statements inquire about your thoughts and feelings in a variety of situations. For each item, indicate how well it describes you by choosing the appropriate one.

	Describes me very well	Describes me well	Describes me somewhat	Does not describe me well	Does not describe me at all
I am often deeply touched by what I see happening to others.	a) (+4)	b) (+3)	c) (+2)	d) (+1)	e) (+0)
I find it easy to see things from other people's points of view.	a) (+4)	b) (+3)	c) (+2)	d) (+1)	e) (+0)
Other people's misfortunes do not usually disturb me a great deal.	a) (+0)	b) (+1)	c) (+2)	d) (+3)	e) (+4)

Note (not shown to participants): High empathy level if score ≥ 10

A5. Donation rates by time discount level

Time discount level	Both	Same day only	In a month only	None	Sum
Low	302 (68.63%)	17 (3.86%)	23 (5.23%)	98 (22.27%)	N=440 (100%)
High	51 (58.62%)	1 (1.15%)	5 (5.75%)	30 (34.48%)	N=87 (100%)

Note: Time discount level is measured as described in A1. A participant is defined as having low time discount if he/she chooses to receive money in one year for all three questions in A1.

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